Product of Array Except Self (Medium)

Problem Statement

Given an array of integers, return a new array where each element at index `i` of the new array is the product of all the numbers in the original array except the one at `i`. You must solve this problem without using division.

Example Generation

1.
   - **Input**: `[2, 3, 4, 5]`
   - **Expected Output**: `[60, 40, 30, 24]`
   - **Justification**: For the first element: `3*4*5 = 60`, for the second element: `2*4*5 = 40`, for the third element: `2*3*5 = 30`, and for the fourth element: `2*3*4 = 24`.
2.
   - **Input**: `[1, 1, 1, 1]`
   - **Expected Output**: `[1, 1, 1, 1]`
   - **Justification**: Every element is 1, so the product of all other numbers for each index is also 1.
3.
   - **Input**: `[10, 20, 30, 40]`
   - **Expected Output**: `[24000, 12000, 8000, 6000]`
   - **Justification**: For the first element: `20*30*40 = 24000`, for the second element: `10*30*40 = 12000`, for the third element: `10*20*40 = 8000`, and for the fourth element: `10*20*30 = 6000`.

Try it yourself

Try solving this question here:

Java

```java
import java.util.Arrays;

public class Solution {
    public int[] productExceptSelf(int[] nums) {
        int[] result = new int[nums.length];
        // ToDo: Write Your Code Here.
        return result;
    }
}

```

Test Case | Run Code Result | Stdout

[2, 3, 4, 5]

Use Example Testcases | Run | Submit

← Previous
Solution: Valid Sudoku

Next →
Solution: Product Of Array Except Self

Mark as Completed